

HAI LI SU, China

WMO: 532310

Lat: 41.4007N

Lon: 106.4070E

Elev: 1510

StdP: 84.45

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-22.3	-32.6	0.2	-17.8	-30.3	0.3	-16.6	12.8	-7.9	11.3	-8.9	3.9	160	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.8	15.0	30.4	14.4	29.0	13.9	18.0	26.1	17.0	25.0	16.1	23.9	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.7	13.4	20.4	14.4	12.3	19.8	13.1	11.3	19.6	57.4	25.9	53.7	24.8	50.7	23.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.2	11.4	10.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-28.2	34.5	2.9	1.8	-30.2	35.8	-31.9	36.8	-33.5	37.8	-35.6	39.1
			-28.4	19.4	2.8	1.4	-30.4	20.4	-32.1	21.3	-33.7	22.1	-35.7	23.1
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-12.7	-7.8	-0.3	8.0	14.4	20.1	23.0	21.0	15.1	6.8	-2.8	-10.6
	DBStd	13.16	4.80	5.50	5.76	5.35	4.58	3.60	2.96	3.28	4.24	5.12	5.43	5.03
	HDD10.0	2795	703	499	322	98	14	0	0	0	9	124	385	640
	HDD18.3	4759	962	733	578	311	135	23	2	12	112	359	635	898
	CDD10.0	1426	0	0	2	39	152	304	403	342	162	24	0	0
	CDD18.3	349	0	0	0	1	14	77	147	96	14	0	0	0
	CDH23.3	3221	0	0	0	11	165	747	1416	783	98	2	0	0
	CDH26.7	982	0	0	0	1	27	211	508	221	15	0	0	0
Wind	WSAvg	5.3	5.1	5.2	5.6	6.0	6.0	5.4	5.0	4.9	5.0	5.1	5.4	5.3
Precipitation	PrecAvg	110	1	1	2	5	8	13	32	27	12	6	2	1
	PrecMax	239	5	3	6	21	29	65	115	96	39	22	15	4
	PrecMin	39	0	0	0	0	0	1	4	3	0	0	0	0
	PrecStd	44	1	1	2	5	8	12	25	21	11	6	3	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.3	8.4	17.1	24.9	29.2	32.5	35.0	32.7	29.0	22.2	12.5	4.3
		MCWB	-2.6	0.5	5.2	9.0	11.5	13.7	17.1	15.7	12.7	9.1	3.1	-1.7
	2%	DB	-0.6	5.2	13.7	22.1	27.1	30.6	32.5	30.8	25.7	19.2	9.6	1.1
		MCWB	-4.8	-1.4	3.2	7.5	10.6	13.1	15.5	14.9	11.5	7.7	1.9	-3.6
	5%	DB	-3.4	2.9	11.0	19.5	25.2	29.1	31.0	29.1	24.1	17.0	7.3	-1.2
		MCWB	-6.6	-2.5	1.9	6.6	9.8	12.8	15.1	14.4	11.2	6.6	0.7	-5.1
	10%	DB	-5.6	0.6	8.6	17.2	23.1	27.6	29.6	27.6	22.3	14.8	5.1	-3.5
		MCWB	-8.2	-3.9	0.6	5.6	8.9	12.3	14.7	14.0	10.7	5.6	-0.6	-6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.3	1.2	5.6	10.1	13.1	17.0	19.8	19.6	15.8	10.4	4.0	-1.4
		MCDB	2.7	7.2	16.1	21.3	23.9	23.4	30.0	24.8	21.4	18.9	11.0	3.8
	2%	WB	-4.7	-0.9	3.6	8.6	11.7	15.4	18.4	18.0	14.3	8.6	2.2	-3.5
		MCDB	-1.0	4.6	12.1	19.4	23.3	23.3	27.5	25.5	20.9	17.7	8.9	0.9
	5%	WB	-6.5	-2.4	2.3	7.3	10.7	14.6	17.2	16.9	13.1	7.2	0.8	-5.0
		MCDB	-3.5	2.5	10.2	18.1	22.4	23.7	25.7	24.9	20.0	15.3	6.9	-1.5
	10%	WB	-8.1	-3.8	1.0	6.0	9.7	13.8	16.4	15.9	11.8	6.1	-0.4	-6.5
		MCDB	-5.6	0.6	8.0	16.6	21.3	23.7	24.6	23.6	19.8	13.9	4.9	-3.5
Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	11.1	11.4	11.6	10.9	10.3	10.2	10.2	10.4	9.9	9.6
		MCDBR	12.1	12.6	13.2	13.4	13.3	12.2	11.6	11.3	11.3	11.8	11.8	11.3
		MCWBR	9.2	8.3	7.4	6.7	6.0	4.3	3.5	3.6	4.2	5.9	7.1	8.1
	5% WB	MCDBR	11.7	11.8	12.0	12.2	11.4	10.0	9.8	9.4	9.3	10.9	11.2	10.9
		MCWBR	9.1	7.9	7.0	6.3	5.7	4.0	3.6	3.7	4.4	5.8	6.9	7.9
Clear-Sky Solar Irradiance	taub	0.283	0.326	0.374	0.425	0.453	0.466	0.449	0.434	0.396	0.353	0.308	0.284	
	taud	2.221	2.115	1.991	1.900	1.889	1.927	2.041	2.062	2.103	2.142	2.191	2.197	
	Ebn at Noon	859	866	861	842	827	815	826	826	830	823	814	819	
	Edh at Noon	94	124	159	187	194	188	166	158	140	119	97	89	
All-Sky Solar Radiation	RadAvg	2.49	3.61	4.99	6.23	7.04	6.95	6.54	6.06	5.11	4.10	2.81	2.21	
	RadStd	0.24	0.20	0.32	0.29	0.30	0.40	0.31	0.26	0.27	0.16	0.18	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page